Rohun Tripathi Contact: rohun.tripathi.5@gmail.com
LinkedIn / Website Address: US

Professional Experience

*Researcher with **10+ years** of experience in **modeling / training** of multimodal AI systems across image, video, robotics and language. Proven track record of developing SoTA models and translating SoTA research into production systems at scale. Experienced in leading research directions and owning end-to-end systems.*

- Research Scientist — Allen Institute for AI (Ai2), US *2023 - Present*
  - Leading model development for SoTA foundation models for robotic reasoning, such as MolmoBot
  - Core contributor for open-source multimodal foundation models (Molmo / Molmo2) for image, video reasoning with SoTA grounding, better than open and proprietary models
- Research Scientist — Amazon Research, US *2018-2023*
  - Led development of multimodal generative systems (GANs, diffusion models) for audio-conditioned video creation in Amazon Prime Video and advertising workflows
  - Developed and deployed few-shot product identification system for Amazon Go and Whole Foods stores, enabling recognition of unseen products in real-world retail environments
  - Led a team of 3 scientists to productize ML systems at scale and deploy them into real-world retail systems
- Research Engineer — IBM Research, India *2017*
- Software Engineer — Sprinklr, India *2015-2017*

Education

Masters in Computer Science | Cornell Tech, Cornell University *2018*
MK Tata Scholarship, 2017

BTech in Computer Science | Indian Institute of Technology, Kanpur *2015*
Academic excellence award for Rank 1, academic year 2013-14
Charpak Scholarship for research semester in France, 2014

Publications

* denotes equal contributors, † denotes core contributors, ‡ denotes equal advising

Peer-Reviewed

- MolmoBoT: Large-Scale Simulation Enables Zero-Shot Manipulation. A Deshpande*†, M Guru*†, R Hendrix*†, S Jauhri*†, A Eftekhar†, R Tripathi†, M Argus†, J Salvador†, H Fang†, M Wallingford†, W Pumacay†, Y Kim†, Q Pfeifer, YC Lee, P Wolters, O Rayyan, M Zhang, J Duan, K Farley, W Han, E Vanderbilt, D Fox, A Farhadi, G Chalvatzaki, D Shah, R Krishna. ICRA 2026 SDRL Workshop (Best paper), ICRA 2026 Beyond Teleop Workshop (Best paper), ICRA 2026 From Data to Decisions Workshop (Best paper). RSS 2026 Sim2Real Workshop. Under Review
- Molmo-Point: Better Pointing for VLMs with Grounding Tokens. C Clark, Y Yang, J S Park, Z Ma, J Zhang, R Tripathi, M Salehi, S Lee, R Krishna. ECCV 2026.
- Molmo2: Open Weights and Data for Vision-Language Models with Video Understanding and Grounding. C Clark*†, J Zhang*†, Z Ma*†, J S Park*†, M Salehi†, R Tripathi†, S Lee†, Z Ren, C D Kim, Y Yang, V Shao, Y Yang, W Huang, Z Gao, T Anderson, J Zhang, J Jain, G Stoica, W Han, A Farhadi, R Krishna. CVPR 2026.
- VideoNet: A Large-Scale Dataset for Domain-Specific Action Recognition. T Yadav, M Salehi, J S Park, V Ramanujan, H Hajishirzi, Y Choi, A Farhadi, R Tripathi‡, R Krishna‡. CVPR 2026.
- SAGE: Training Smart Any-Horizon Agents for Long Video Reasoning with Reinforcement Learning. J Jain, J Li, Z Ma, J Zhang, C D Kim, S Lee, R Tripathi, T Gupta, C Clark, H Shi. CVPR 2026.
- HinTel-AlignBench: A Framework and Benchmark for Hindi-Telugu with English-Aligned Samples. R Chigrupaatii, P Kanishka, L Routhu, M Patel S Reddy, D Gupta, D Srikar, K Kuchimanchi, R Misra, R Tripathi. ACL 2026 AVLR Workshop.
- Molmo and PixMo: Open Weights and Open Data for State-of-the-Art Vision-Language Models. M Deitke*, C Clark*, S Lee, R Tripathi and others. CVPR 2025.
  Best paper Honorable Mention - awarded to top 6 papers of 13K submissions
  Oral - awarded to top 0.7% of 13K submissions
- SIDGAN: High-Resolution Dubbed Video Generation via Shift-Invariant Learning. U Muaz*, W Jang*, R Tripathi, S Mani, W Ouyang, R T Gadde, B Gecer, S Elizondo, R Madad, N Nair. ICCV 2023.

- ModEFormer: Modality-preserving embedding for audio-video synchronization using transformers. A Gupta, R Tripathi, W Jang. ICASSP 2023.
- Automatic Generation of Secure and Usable Audio reCAPTCHA. R Tripathi*, M Jain*, P Kumar, S Patel. ASSETS 2019.
- Enterprise scale privacy aware occupancy sensing. S S K Sajja, A P K S Prakash, R Tripathi, S Dwivedi, A Singhee, M Vermeulen. IEEE EDGE 2018.
- Semantic Segmentation with Scarce Data. R Tripathi*, I Katsman*, A Veit, S Belongie. ICML TLLL Workshop 2018.

Pre-prints

- DCVLR: Data Curation for Vision Language Reasoning Competition. O Elachqar*, B Feuer*, N Hulkund*, R Tripathi*, R Agrawal, T Nguyen, N Shabtay, V Udandarao, X Wang, S Webb, Y Zhang, S Beery, G Gkioxari, P Liang, L Schmidt, S Xie, S Yeung-Levy, M Buriek, P Bushuyeu, D Damianos, E Kasoura, V Katsouros, V Kumar, I Molybog, G Paraskevopoulos, Y Shin, B Sobolev, S Watson, H Xu, E Koukoumidis. Under Review
- Unified Spatio-Temporal Token Scoring for Efficient Video-Language Models. J Zhang, Y Yang, R Tripathi, W Han, R Krishna, C Clark, Y J Lee, S Lee. Under Review
- RSO: A Gradient Free Sampling Based Approach For Training Deep Neural Networks. R Tripathi, B Singh. Arxiv, 2020
- ASAP-NMS: Accelerating Non-Maximum Suppression Using Spatially Aware Priors. R Tripathi*, V Singla*, M Najibi, B Singh, A Sharma, L Davis. Arxiv, 2020
- Single-View Height Map Prediction for Satellite Imagery. R Tripathi, N Snavely.

Honors and Awards

- MolmoBoT: Large-Scale Simulation Enables Zero-Shot Manipulation.
  - Best Paper Award at the SDRL Workshop ICRA 2026
  - Best Paper Award at the Beyond Teleoperation Workshop ICRA 2026
  - Best Paper Award at the From Data to Decisions Workshop ICRA 2026
- Molmo2: Open Weights and Data for Vision-Language Models with Video Understanding and Grounding (CVPR 2026)
  - Best Paper Nominee at CVPR (Top 0.5% of submissions)
  - Oral Presentation at CVPR (Top 0.9% of submissions)
- VideoNet: A Large-Scale Dataset for Domain-Specific Action Recognition (CVPR 2026)
  - Highlight Presentation at CVPR (Top 3.6% of submissions)
  - Best Paper Award (Benchmark Track) at the KnowledgeMR Workshop, CVPR
- Molmo and PixMo: Open Weights and Open Data for State-of-the-Art Vision-Language Models (CVPR 2025)
  - Best Paper Honorable Mention at CVPR (Runner-Up to Best Paper Award)
  - Oral Presentation at CVPR (Top 0.7% of submissions)
- MK Tata Scholarship to pursue graduate studies at Cornell Tech, 2017
- Academic excellence award for Rank 1 at IIT Kanpur, Academic Year 2013-14
- Charpak Scholarship for study abroad semester in France, 2014

Academic Service

Workshop Organization: Lead Co-organizer for Data Curation for VLM reasoning Competition Workshop, Neurips 2025
Paper Review: CVPR 2024, 25, 26; ICML 26; ACL ARR 26; Neurips 26; NeurIPS W 2024; BMVC 26; AMLC 2020, 22

Patents

- Performance characteristic transfer for localized content. *R Tripathi*, A Saha, N S Nair. Granted.
- Systems and methods for automatic aspect ratio adjustment for video content. *R Tripathi*, B Gecer. Granted.

Press Coverages

- Ai2 Unveils Molmobot And Molmospaces For Zero-Shot Robot Training In Simulation The AI Economy, AINews, menafn, Ai.cc
- Ai2's Molmo 2 shows open-source models can rival proprietary giants in video understanding VentureBeat, GeekWire, The Outpost, The AI Economy
- Ai2's new Molmo open source AI models beat GPT-4o, Claude on some benchmarks. VentureBeat, MIT Tenchnology, Wired, TechCrunch
- Amazon's new generative AI tool lets advertisers enhance product images. TechCrunch, PYMNTS, About Amazon
- Whole Foods / Amazon Fresh Stores with Just Walk Out Technology. Business Insider, PYMNTS, GeekWire, Verge

Invited Talks

Molmo (Allen Institute-1, Jun. 2025); Molmo2 (Amazon Prime Video, Feb. 2026; DLCT, Apr. 2026)

Selected Relevant Projects

- **Conditional Video Generation, Amazon Prime Video** *2023*
  - Developed video GANs, Diffusion Models and NeRFs, with generation driven by audio or text.
  - Developed audio-video synchronization models with SoTA performance.
- **Few-shot Product Identification, Amazon Go** *2021*
  - Developed scene understanding models and angle-aware product identification for unseen classes
  - Led team of 3 scientists to deploy both models in Go stores and select Whole Foods stores.
- **Gradient-Free Optimization of Neural Networks** *2019*
  - Trained a deep network without gradient based optimization by perturbing the weights sequentially.
  - Outperformed ES and on par with SGD for image classification on CIFAR and MNIST.

Mentorship

- Ph.D. students (placement after mentorship)
  - Yuk Heo (continued at Korea University), 2022
- Masters students (placement after mentorship)
  - Akash Gupta (placement: Robotics and AI Institute), 2022
  - Rohan Agarwal (placement: Runway Research), 2022
- Undergraduate students (placement after mentorship)
  - Tanush Yadav (now PhD student at University of Washington), 2025
  - Rishikant Chigrupaatii (placement: Sprinklr), 2025
  - Lalit Chandra (placement: Warner Brothers Discovery), 2025
  - Martin Patel, 2025
  - Ponnada Sai Tulasi Kanishka, 2025